Euclidean Algorithm and Bézout's Identity

An End-to-End Meta-Math Walkthrough

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Abstract

A decreasing remainder sequence proves termination; a second invariant carries a certificate in the original signed inputs. Their convergence identifies the positive gcd and supplies modular inverses. This report presents a complete proof relative to explicit imports, a dependency graph, versioned node contracts, five-component activities, exact examples, and an illustrative repair. The interactions are authored explanations, not a transcript of deployed agents or a proof-assistant certificate.

Project NT-01 Report version 1 Number Theory

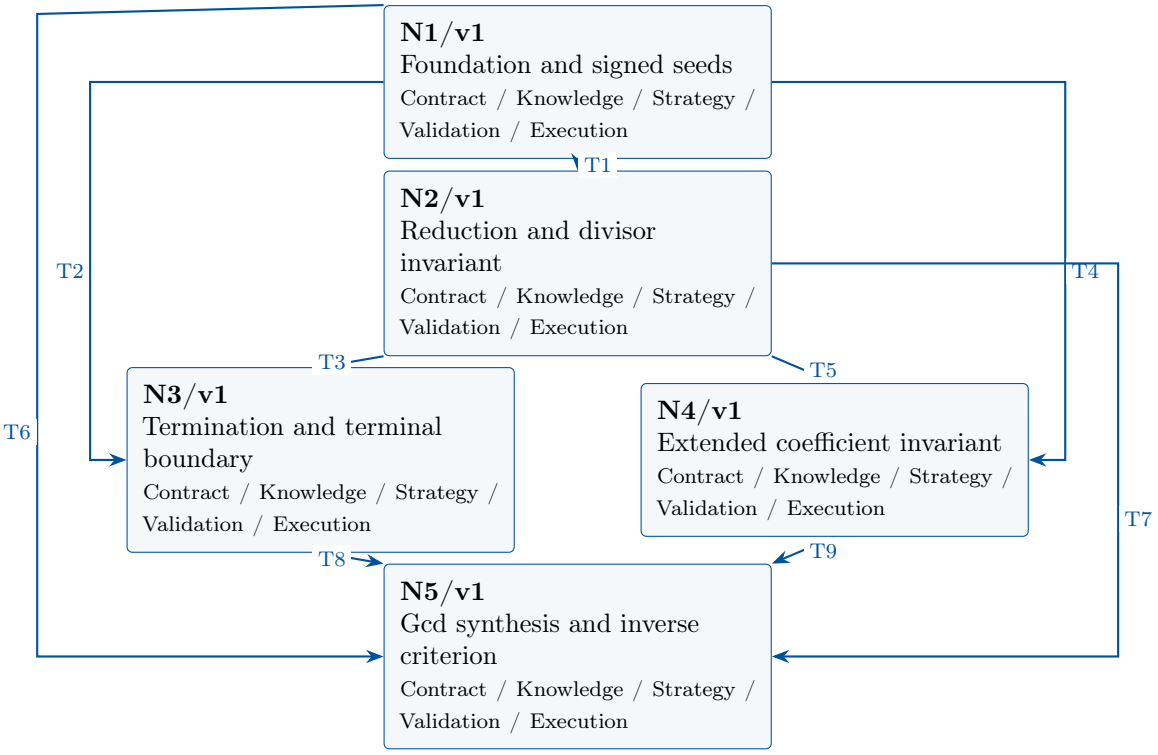
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For every $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$, the extended Euclidean algorithm terminates and returns integers d, u, v such that $d = \gcd(a, b) > 0$ and $au + bv = d$. For every $m \geq 2$, an integer a is invertible modulo m if and only if $\gcd(a, m) = 1$.*

A. Foundations and permitted imports

1. Divisibility means $c \mid x$ if $x = ck$ for an integer k. The gcd is the largest positive common divisor; positive common divisors form a nonempty finite set when the inputs are not both zero.
2. Division with remainder: for $A \geq 0$ and $B > 0$, unique integers q, r satisfy $A = qB + r$ and $0 \leq r < B$. Well-ordering rules out an infinite strictly decreasing sequence of positive integers.
3. Congruence $x \equiv y \pmod{m}$ means $m \mid (x - y)$. No Bézout theorem, Euclid lemma, or unique factorization is imported.

II. NODE INTERACTION WORKFLOW



Blue arrows are accepted mathematical dependencies, with exact transfers T1–T9 below. Repair messages are described separately; they are not premises of the proof.

A. Execution loop associated with the proof

The repeated computation is an execution loop, not a cycle of proof premises. All right-hand sides in the final assignment use the old state.

```
require integer a, b, not both zero
A, B = abs(a), abs(b)
sA, tA = sign(a), 0
sB, tB = 0, sign(b)
while B != 0:
    q, r = divmod(A, B)
    nextS, nextT = sA - q*sB, tA - q*tB
    assert 0 <= r < B and A == q*B + r
    assert r == nextS*a + nextT*b
    A, B, sA, tA, sB, tB = B, r, sB, tB, nextS, nextT
return A, sA, tA
```

B. Versioned evidence-transfer ledger

T1: N1/E1 v1 to N2.1
Claim. $r_0 = |a|$, $r_1 = |b|$ have the same common divisors as a, b; $r_i = s_i a + t_i b$ for $i = 0, 1$.
Compatibility. Normalized nonnegative pair; division still requires a positive second entry.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T2: N1/E1 v1 to N3.3
Claim. $r_0 = |a|$, $r_1 = |b|$ have the same common divisors as a, b; $r_i = s_i a + t_i b$ for $i = 0, 1$.
Compatibility. The original pair is not both zero.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T3: N2/E2 v1 to N3.1
Claim. $A = qB + r$, $0 \le r < B$, and the pairs (A, B) and (B, r) have identical positive common divisors.
Compatibility. The next divisor is strictly smaller whenever the run continues.
Evidence. Node N2 activities below and the continuous proof in Section IV.

T4: N1/E1 v1 to N4.3
Claim. $r_0 = |a|$, $r_1 = |b|$ have the same common divisors as a, b; $r_i = s_i a + t_i b$ for $i = 0, 1$.
Compatibility. Both seed coefficient identities hold.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T5: N2/E2 v1 to N4.1
Claim. $A = qB + r$, $0 \le r < B$, and the pairs (A, B) and (B, r) have identical positive common divisors.
Compatibility. Use the very same quotients as the remainder run.
Evidence. Node N2 activities below and the continuous proof in Section IV.

T6: N1/E1 v1 to N5.1
Claim. $r_0 = |a|$, $r_1 = |b|$ have the same common divisors as a, b; $r_i = s_i a + t_i b$ for $i = 0, 1$.
Compatibility. The original signed inputs have the normalized divisor set.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T7: N2/E2 v1 to N5.1
Claim. $A = qB + r$, $0 \le r < B$, and the pairs (A, B) and (B, r) have identical positive common divisors.
Compatibility. Divisor-set preservation holds through every valid step.
Evidence. Node N2 activities below and the continuous proof in Section IV.

T8: N3/E3 v1 to N5.1
Claim. Every valid run reaches $(d, 0)$ after finitely many divisions, with $d > 0$.
Compatibility. A finite positive terminal remainder exists.
Evidence. Node N3 activities below and the continuous proof in Section IV.

T9: N4/E4 v1 to N5.2
Claim. Every generated remainder satisfies $r_j = s_j a + t_j b$ with integer coefficients.
Compatibility. Select coefficients attached to the last nonzero remainder.
Evidence. Node N4 activities below and the continuous proof in Section IV.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Foundation and signed seeds

Object and role. Own the input domain and two initial coefficient identities; enable later certificates to refer to the original inputs.

Inputs. $a, b \in \mathbb{Z}$, not both zero.

C — Contract

Publish only admissible inputs and signed seed identities; keep the target gcd positive.

K — Knowledge

Retain the input pair, divisibility, absolute value, and $\varepsilon(x) = \operatorname{sgn}(x)$ with $\varepsilon(0) = 0$.

S — Strategy

Normalize magnitudes while expressing coefficients in the original signed integers.

V — Validation

Reject $(0, 0)$; permit one zero and either input order.

E — Execution

Set $(r_0, r_1) = (|a|, |b|)$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N1.1 Set $(r_0, r_1) = (|a|, |b|)$.

N1.2 Set $(s_0, t_0) = (\varepsilon(a), 0)$ and $(s_1, t_1) = (0, \varepsilon(b))$.

N1.3 Check $r_0 = s_0 a + t_0 b$ and $r_1 = s_1 a + t_1 b$; $c \mid x$ iff $c \mid |x|$.

Export E1/v1. $r_0 = |a|$, $r_1 = |b|$ have the same common divisors as a, b; $r_i = s_i a + t_i b$ for $i = 0, 1$.

Acceptance gate. Both seed identities hold and at least one remainder is positive.

Targeted repair. If an input sign changes, regenerate its seed and recheck N4 and N5.

B. N2/v1: Reduction and divisor invariant

Object and role. Own a valid division and the equality of common-divisor sets; supply both progress and coefficient updates.

Inputs. N1/E1; a current pair $A \geq 0$, $B > 0$.

C — Contract

Publish preservation for a valid step, without asserting that a complete run has terminated.

K — Knowledge

Retrieve division with remainder and closure of divisibility under integer linear combinations.

S — Strategy

Prove both containments of the common-divisor sets.

V — Validation

Require $B > 0$ before division and audit the strict remainder bound.

E — Execution

Compute $A = qB + r$ with $0 \leq r < B$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N2.1 Compute $A = qB + r$ with $0 \leq r < B$.

N2.2 If $c \mid A, B$, then $c \mid A - qB = r$.

N2.3 If $c \mid B, r$, then $c \mid qB + r = A$.

Export E2/v1. $A = qB + r$, $0 \leq r < B$, and the pairs (A, B) and (B, r) have identical positive common divisors.

Acceptance gate. The division equation, remainder bounds, and both divisor implications are established.

Targeted repair. An inadmissible zero divisor routes to the stop condition, not to a fabricated next remainder; recheck N3–N5.

C. N3/v1: Termination and terminal boundary

Object and role. Own the decreasing integer measure and the positive terminal value.

Inputs. N2/E2 and the nonzero-input condition in N1/E1.

C — Contract

Export finite termination and positivity, leaving gcd identification to N5.

K — Knowledge

Retain well-ordering and all remainder inequalities from N2.

S — Strategy

Use the current positive divisor as the strictly decreasing measure.

V — Validation

Stop when the second entry is zero, including the initial state; do not require the first quotient to be positive.

E — Execution

While $r_j > 0$, form $0 \leq r_{j+1} < r_j$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N3.1 While $r_j > 0$, form $0 \leq r_{j+1} < r_j$.

N3.2 An infinite run would be an infinite descent of positive integers.

N3.3 At termination the first entry is positive: it was the last positive divisor, or the nonzero initial magnitude.

Export E3/v1. Every valid run reaches $(d, 0)$ after finitely many divisions, with $d > 0$.

Acceptance gate. Both immediate termination and a run with divisions end at a positive first entry and zero second entry.

Targeted repair. Replace an incorrect continuation test by $B \neq 0$; invalidate a proposed extra transition and recheck N5.

D. N4/v1: Extended coefficient invariant

Object and role. Transport each remainder reduction into integer coefficient space.

Inputs. N1/E1 seeds and the quotient sequence from N2/E2.

C — Contract

Export an identity for each remainder; do not identify a gcd from an identity alone.

K — Knowledge

Retain both preceding coefficient pairs and their matching remainder indices.

S — Strategy

Apply exactly the same linear recurrence to remainders and their coefficients.

V — Validation

Use the old pairs simultaneously; distinguish the coefficients of the last nonzero remainder from those of zero.

E — Execution

Define $s_{j+1} = s_{j-1} - q_j s_j$ and $t_{j+1} = t_{j-1} - q_j t_j$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N4.1 Define $s_{j+1} = s_{j-1} - q_j s_j$ and $t_{j+1} = t_{j-1} - q_j t_j$.

N4.2 Substitute the two preceding identities into $r_{j+1} = r_{j-1} - q_j r_j$.

N4.3 Induct from the seed identities to every generated index.

Export E4/v1. Every generated remainder satisfies $r_j = s_j a + t_j b$ with integer coefficients.

Acceptance gate. The recurrence preserves $r_{j+1} = s_{j+1} a + t_{j+1} b$ at every step.

Targeted repair. Recompute the first mismatched coefficient pair; downstream pairs and N5 must be rechecked.

E. N5/v1: Gcd synthesis and inverse criterion

Object and role. Join divisor preservation, termination, and the coefficient invariant; export both directions of the inverse criterion.

Inputs. N1–N4 for the same input pair and remainder sequence.

C — Contract

Publish the theorem only after positivity, divisibility, maximality, and the signed certificate are joined.

K — Knowledge

Retrieve the terminal pair, the preserved divisor set, its coefficient pair, and the definition of congruence.

S — Strategy

Identify the divisors of $(d, 0)$ and turn a Bézout identity into a modular inverse.

V — Validation

A combination alone does not identify a gcd; check both directions of the inverse equivalence.

E — Execution

The common divisors of the original pair equal the positive divisors of d; thus $d \mid a, b$ and any common divisor is at most d. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N5.1 The common divisors of the original pair equal the positive divisors of d; thus $d \mid a, b$ and any common divisor is at most d.

N5.2 Take (u, v) from the coefficient pair belonging to d.

N5.3 If $au + mv = 1$, reduce modulo m; conversely $aw \equiv 1 \pmod{m}$ gives $aw - km = 1$.

Export E5/v1. $d = \gcd(a, b) > 0$, $au + bv = d$; a is a unit modulo $m \geq 2$ iff $\gcd(a, m) = 1$.

Acceptance gate. The gcd contract and both inverse implications close without division by a nonunit.

Targeted repair. A failed final certificate returns to N4; a failed divisor conclusion returns to N2. Rejoin only corrected exports.

IV. CONTINUOUS MATHEMATICAL PROOF

Proof. Let $r_0 = |a|$, $r_1 = |b|$, and initialize the signed coefficient pairs as in N1. If $r_1 = 0$, then $r_0 > 0$ and the terminal state is already available. Otherwise, repeatedly divide to obtain

$$r_{j-1} = q_j r_j + r_{j+1}, \qquad 0 \le r_{j+1} < r_j.$$

For any positive integer c, subtraction and addition show that $c \mid r_{j-1}, r_j$ if and only if $c \mid r_j, r_{j+1}$ (N2). Thus the common-divisor set is invariant. The positive divisors used in successive divisions strictly decrease, so well-ordering forces termination (N3) at $(d, 0)$ with $d > 0$.

The terminal positive common divisors are exactly the positive divisors of d. Preservation therefore implies $d \mid a, b$, and every positive common divisor c of a, b divides d, hence $c \le d$. This proves $d = \gcd(a, b)$.

Define the coefficient recurrence from N4. Assuming the two preceding coefficient identities,

$$\begin{aligned} r_{j+1} &= r_{j-1} - q_j r_j \\ &= (s_{j-1} - q_j s_j)a + (t_{j-1} - q_j t_j)b. \end{aligned}$$

The initial identities and induction establish the invariant at every index. The coefficients belonging to d therefore give integers u, v with $au + bv = d$; the immediate-stop case uses the same seeds.

If $\gcd(a, m) = 1$, the result gives $au + mv = 1$, so u is an inverse modulo m. Conversely an inverse w gives $aw - 1 = km$ for an integer k. Every common divisor of a, m divides $aw - km = 1$, forcing their positive gcd to be 1. This proves both claims (N5).

□

V. WORKED ORCHESTRATION

For $(a, b) = (252, 198)$, the exact remainder and coefficient trace is

Remainder	Division	Coefficients (s, t)
252	Initial state	$(1, 0)$
198	Initial state	$(0, 1)$
54	$252 = 198 + 54$	$(1, -1)$
36	$198 = 3 \cdot 54 + 36$	$(-3, 4)$
18	$54 = 36 + 18$	$(4, -5)$
0	$36 = 2 \cdot 18$	$(-11, 14)$

N2 produces each division; N3 checks descent; N4 updates the matching coefficient pair. N5 uses the last **nonzero** row: $18 = 4 \cdot 252 - 5 \cdot 198 = 1008 - 990$, with $252 = 14 \cdot 18$ and $198 = 11 \cdot 18$.

For an inverse, divisions $43 = 2 \cdot 17 + 9$, $17 = 9 + 8$, $9 = 8 + 1$ give

$$1 = 9 - 8 = 2 \cdot 9 - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Thus $17^{-1} \equiv -5 \equiv 38 \pmod{43}$, checked by $17 \cdot 38 = 15 \cdot 43 + 1$. If the algorithm starts with $(17, 43)$, its first quotient is simply zero.

A. Boundary and hypothesis audit

- $(0, 0)$ is excluded. For $(a, 0)$ with $a \neq 0$, return $(|a|, \operatorname{sgn}(a), 0)$; for $(0, b)$ one valid division gives $(|b|, 0, \operatorname{sgn}(b))$.
- For $(-252, 198)$, the signed certificate is $18 = (-4)(-252) - 5 \cdot 198$. Bézout coefficients need not be unique.
- 6 has no inverse modulo 15: every expression $6w - 15k$ is divisible by 3, so it cannot equal 1.

VI. ILLUSTRATIVE FAILURE AND REPAIR

This is a deliberately authored teaching trace. A diagnostic asks for correction; it cannot prove a predecessor by importing its descendant's conclusion. A revised export must be checked before its consumers are reused.

Candidate. At the valid terminal state $(18, 0)$, a controller requests another division because the first entry is positive.

Detection. N2 Validation blocks division by zero; N3 Knowledge supplies the stopping condition on the second entry.

Correction. Strategy changes the guard to $Be0$. Execution replays $(36, 18)o(18, 0)$ and selects coefficients $(4, -5)$.

Recheck. Discard the proposed extra transition; recheck the terminal certificate in N4 and the synthesis in N5.

Publication. Contract keeps the original theorem. A hypothetical controller revision is v2; the accepted authored mathematical exports remain v1.

VII. REUSABLE RESULT AND PROVENANCE

Reusable exports retain each node's version 1 statement, assumptions, and proof. The final export is the theorem in Section 1, proved relative to its declared imports.

Source record. Adapted from the original project Markdown:
`euclidean-algorithm-and-bezout-identity.md`

Typography, workflow conventions, and component cards follow the workspace Meta-Math manuscript. This authored exposition does not claim formal certification.